

Homework 3

Due September 16, 2025.

Problem 1. Q_{2n}

Consider the dicyclic group Q_{2n} whose presentation is as follows

$$Q_{2n} = \{a, b \mid a^{2n} = e, a^n = b^2, bab^{-1} = a^{-1}\} \quad (1)$$

In class we discussed the quaternion group Q_4 of order 8; now we consider Q_6 of order 12.

- What are the group elements of Q_6 ? What is their order? Do all group elements in a given class have the same order? If $ba^k = a^l b$ what are k and l ? What are the classes of Q_6 ? Draw a circle around each class, and write the order of each class under the class.
- Q_6 has 3 proper normal subgroups; what are they? What is the commutator subgroup $C(Q_6)$?
- Show that Q_6 can be written as the semidirect product $Z_3 \rtimes Z_4$. (*Hint*: $Z_4 = \{e, b, b^2, b^3\}$.)
- What is the center $Z(Q_6)$ of Q_6 ? What is the centralizer Z_b of b ?
- Prove that the following is a representation (a realization in terms of matrices) of the group Q_6

$$a = \frac{1}{2}(I + i\sqrt{3}\sigma_3); \quad b = i\sigma_2 \quad (\text{or } b = i\sigma_1), \quad (2)$$

where I is the 2×2 unit matrix and σ_i are the Pauli matrices.

- As we shall prove in class, there are as many irreducible representations of a finite group as there are classes. If d_{R_i} denotes the dimension of the representation R_i , then $|G| = \sum_{i=1}^{n_C} (d_{R_i})^2$ where n_C is the number of classes. The number of 1-dimensional representation is $|G|/C(G)$. Use these theorems to deduce the dimensions of the irreducible representation of Q_6 .

Problem 2. A_5

The group A_5 is famous for two reasons: it is the first simple group after the cyclic groups of prime order, and it is the rotation group of an icosahedron. We shall study both properties.

- What is the order of A_5 ? What are the classes of A_5 ? (Recall that the 5-cycles form two classes.) What is the order of these classes? If A_5 were not simple, it would have a proper normal subgroup. A proper normal subgroup consists of entire classes. Show that no combination of classes yields a subgroup. (Use that the order of a subgroup is a divisor of the order of the group.) This proves that A_5 is simple. [It is claimed that Evariste Galois

was the first to prove that A_5 is simple. (His proof was lost or never recorded.) He used this property for his proof that not all quintic polynomial equations have only radical roots.]

- b) Consider an icosahedron. Show that the number of vertices V , edges E , and faces F , satisfies Euler's equation $V - F + E = 2$. How many rotations are there around an axis that connects two opposite vertices? And about an axis through the middle of two opposite faces? (We call such an axis a crossline.) And through the middle of two opposite edges?
- c) Now we are going to show that these rotations correspond to the permutations of the alternating group A_5 . To show this we consider the 15 crosslines and divide them into 5 triplets. A triplet is constructed as follows. Put an icosahedron in front of you on a table such that it is resting on an edge (you must keep it upright). The vertical crossline is the first of the triplet we will construct. Next consider a horizontal plane through the center of the icosahedron. There are two crosslines in this plane; together with the vertical crossline this constitutes one of the five triplets. Draw a picture of the intersection of the horizontal plane with the icosahedron, and draw in this picture the two orthogonal crosslines. [We will put on reserve in the Math-Phys library the icosahedron shown in class. You can ask for it to study it, but you cannot check it out.]
- d) Each rotation corresponds to an even permutation of the five triplets. Identify the rotations established before with the group elements of A_5 . This proves that A_5 is the rotation group of A_5 .
- e) What is the full isometry group of an icosahedron (rotations and reflections)? Is it S_5 ? Or $A_5 \times (e, i)$ where i is space inversion? Or something else? [Felix Klein, of the Klein group V , studied the icosahedron, in his "Erlanger program" of 1872 in which he proposed to classify and unify different geometries by focussing on their group properties.]